Predicting Severe Asthma Exacerbations

A Genome-Wide Association Study using **Random Forests** Classifiers

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Dec 2025 | GTIIT's 3rd International Symposium Presentation

Self Introduction

Mousheng Xu, PhD

Head of Biosciences, A Private Company In Singapore

- **Academic Foundation:**
  - B.S. in Genetics (Wuhan University)
  - M.S. in Biochemistry & Molecular Biology, M.S. in Computer Science (University of Kansas)
  - PhD in Bioinformatics (Boston University; Harvard Medical School)
  - Postdoc in Cancer Genetic Epidemiology (Harvard T.H. Chan School of Public Health).
- **20+ Years Experience:** Software development, computational biology, statistical genetics, and machine learning.
- **Key Achievements:**
  - The Encyclopedia of DNA Elements (ENCODE)
  - Million Veteran Program (MVP)
  - Founded Hubei Vidagen (Genetic Testing)
  - Scientific Breakthrough Of the Year (ATS 2009)

The Clinical Challenge：Childhood Severe Asthma Exacerbations

Definition

Severe Asthma Exacerbations: **emergency room visit or a hospitalization** for asthma symptoms.

Prognostic Significance

Earlier therapeutic interventions.

The Clinical Challenge

- **Prediction with clinical traits is not good enough**
- **Genomic information might be helpful for prediction**

The Problem: "Small N, Large P" Problem

Sample Size (N)

127 cases

290 controls

417 total

Predictors (P)

> 550,000

Single Nucleotide Polymorphisms (SNPs) + Clinical Traits

Standard Method

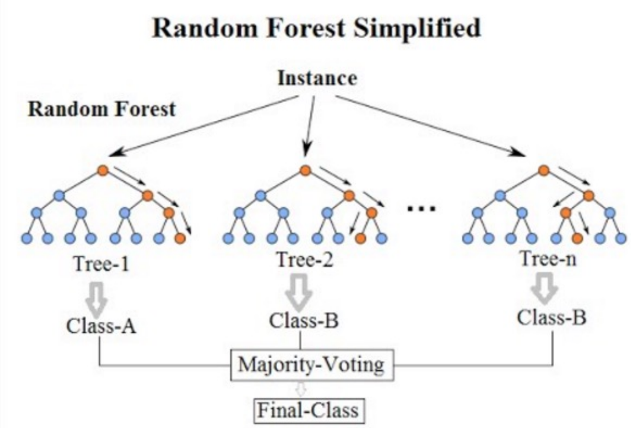
Logistic Regression

Struggles when predictors (P) vastly outnumber samples (N)

Problem: overfitting

Our Solution: Random Forests

Leveraging ensemble power for robust predictive modeling.

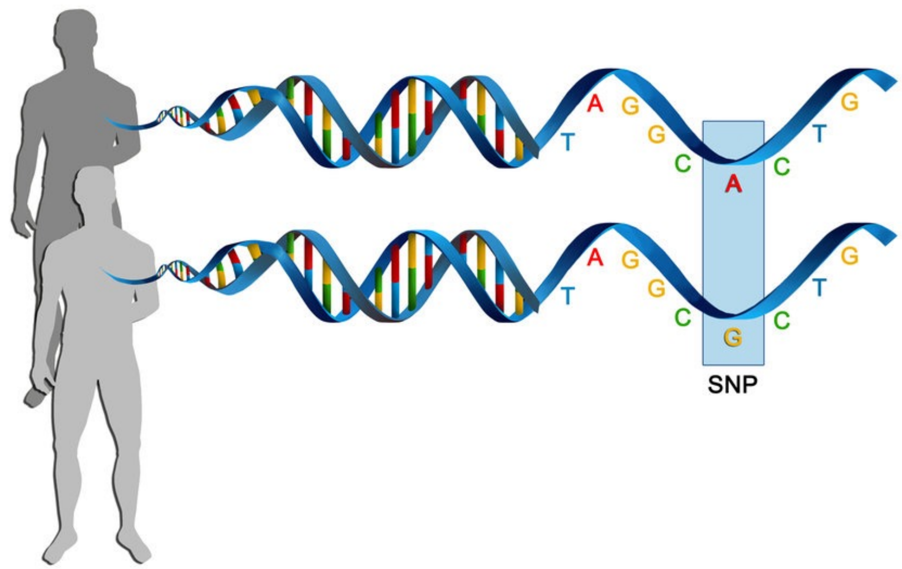


Key Concepts of Random Forests

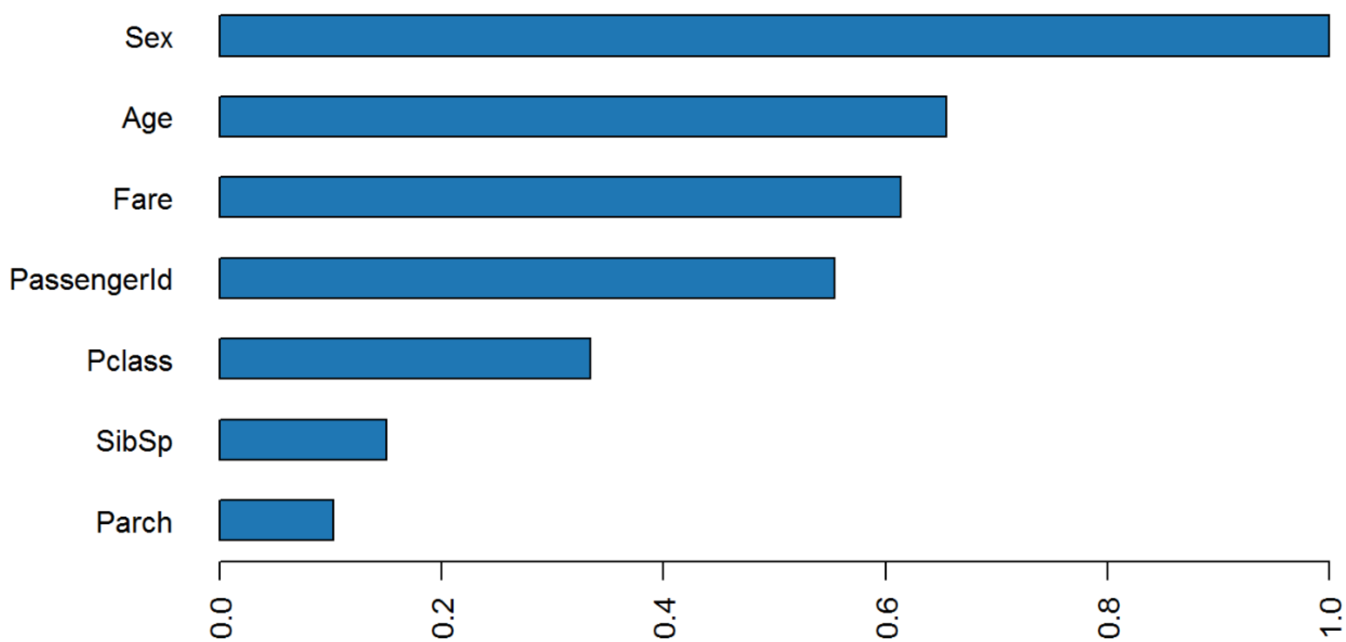
- A **decision tree** predicts outcomes by splitting data based on feature values.
- An **ensemble of many decision trees** forms a "forest".
- Each tree uses **random samples** of the data and considers only **random features** at each split, enhancing diversity.
- **Voting Mechanism:** The model aggregates predictions by taking the **votes from each decision tree** to determine the final outcome.
- Handles **nonlinear, high-dimensional data** effectively, crucial for complex datasets.
- It is known to be relatively resistant to overfitting

Key Terminology： SNP

- **<u>S</u>ingle <u>N</u>ucleotide <u>P</u>olymorphism**
- A one-letter difference (A/C/G/T) on DNA
- May influence traits or disease risks



Key Terminology: RF Importance Score



- Random Forest **predictive power**: contribution to the predictive accuracy
- The higher, the more contribution

Key Terminology: AUC

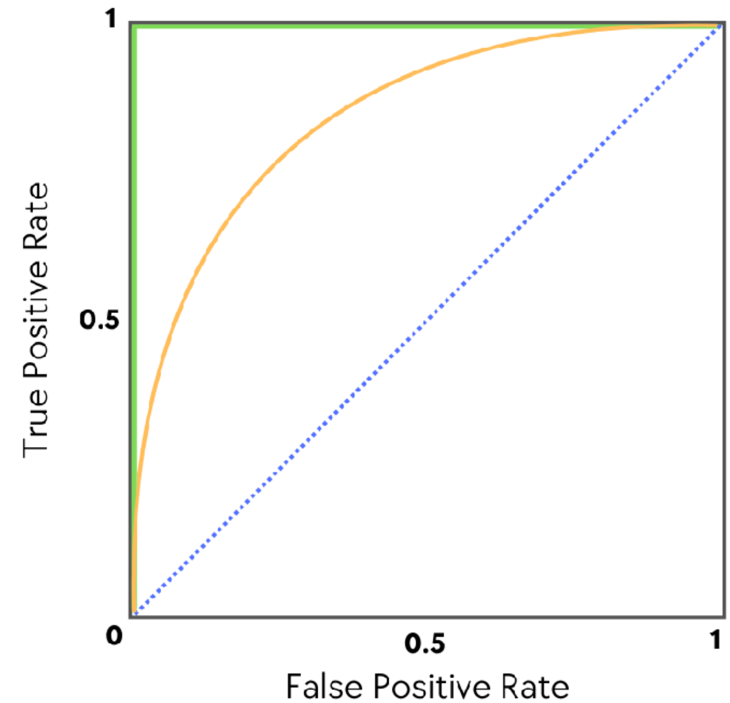
		True Class: T	True Class: F
Acquired Class	Y	True Positives (TP)	False Positives (FP)
	N	False Negatives (FN)	True Negatives (TN)

$$\text{True Positive Rate (TPR)} = \frac{TP}{TP + FN}$$

$$\text{False Positive Rate (FPR)} = \frac{FP}{FP + TN}$$

$$\text{Accuracy (ACC)} = \frac{TP + TN}{TP + FP + TN + FN}$$

- **Area Under the (ROC) Curve**
- A performance metric ranging from 0 to 1
- Measures the ability to distinguish between classes
- **0.5 is random guessing**
- **1.0 is perfect prediction**
- **Receiver-Operating Characteristic (ROC) curve**



Methodology: Feature (SNP) Selection

Step 1: Initial Screen

RF RF RF

4k SNPs 4k SNPs … 4k SNPs

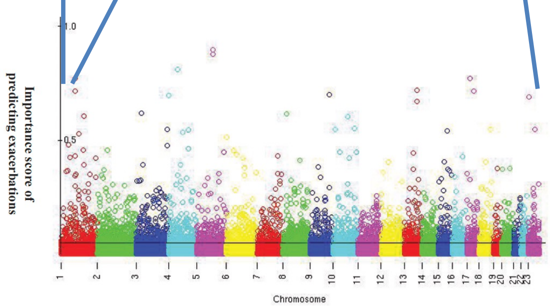


Figure 1 The "manhattan plot" of RF importance scores of all the 550k SNPs. X-axis: the SNPs in chromosomal order; Y-axis: the RF importance scores. The black demarcation separates the top 4k SNPs from the rest.

Step 2: Ranking

Top 4k SNPs Importance Score

RF

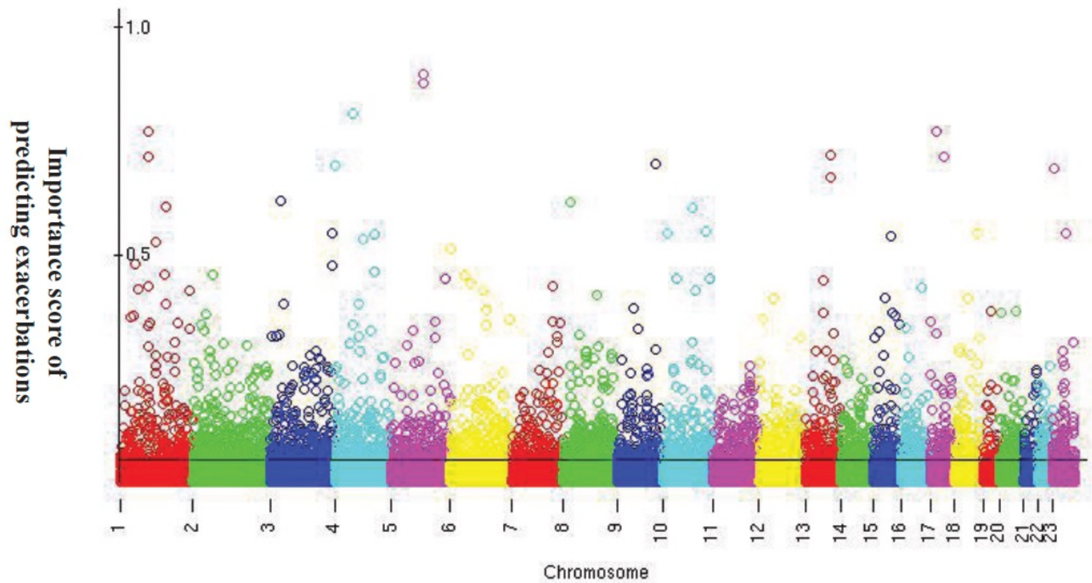
Reranked 4k SNPs Importance Score

Step 3: Prediction

RF

Build final models using Top N SNPs (10, 40, 160, 320) + 4 Clinical Traits.

The RF Importance Landscape



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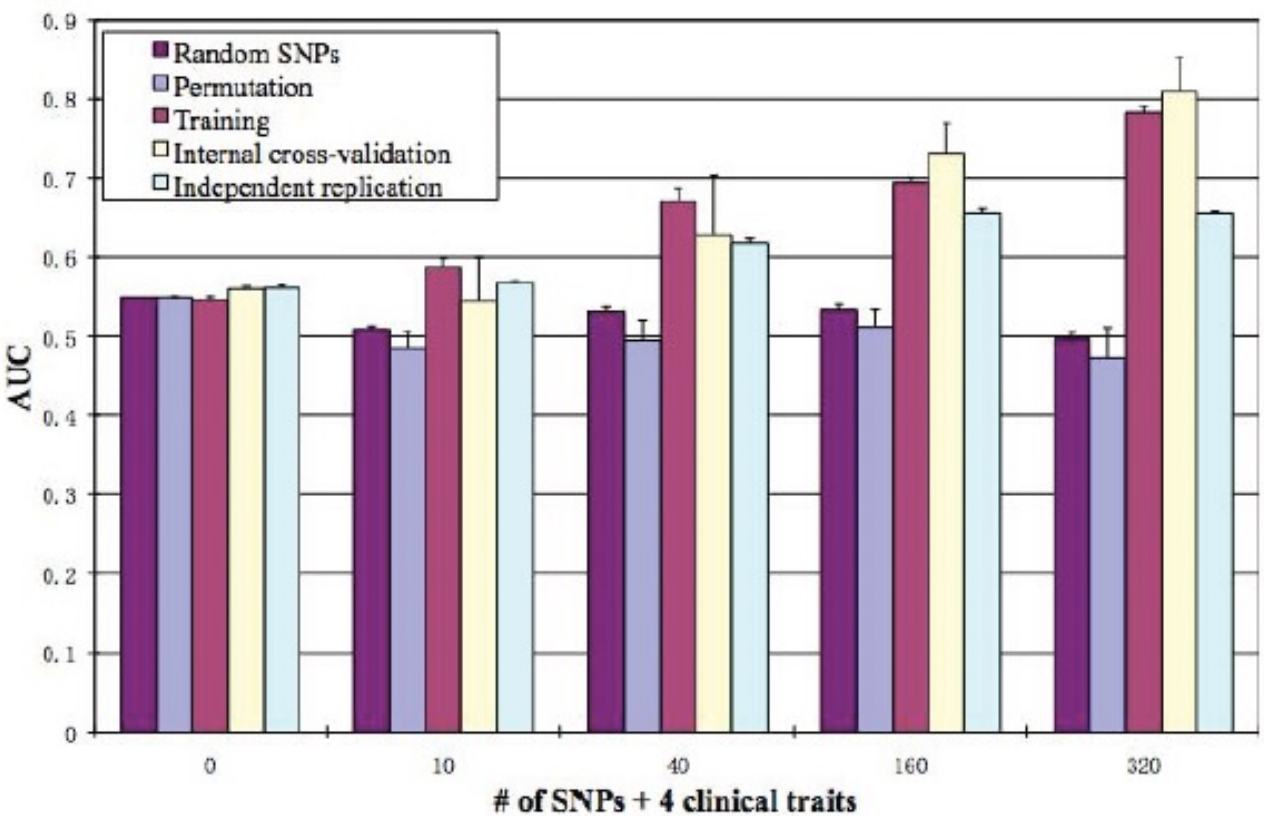
Predictive Power

Area Under the Curve (AUC) in Independent Replication Cohort



- Adding genetic markers significantly improved prediction over clinical history alone.
- The model plateaued at 160 SNPs
- **Permutation Testing:** Confirmed results were not due to chance.
- **Independent Replication:** The model was tested on a separate cohort of 164 subjects, maintaining an AUC of 0.66.

Top Genes

95	94	rs2659695	GATA2	5UP	0.030545646
96	95	rs10745641	NUDT4	5UP	0.029515357
97	96	rs2206860	DNAH8	INTRON	0.029494229
98	97	rs6862273	ADCY2	5UP	0.029459742
99	98	rs2046626	TBC1D9	INTRON	0.029255093
100	99	rs2512033	MATN2	INTRON	0.028883171
101	100	rs10496476	DPP10	INTRON	0.028850932
102	101	rs616246	PFKFB3	5UP	0.028848902
103	102	rs12424252	XRCC6BP1	3DOWN	0.02878733
104	103	rs2109818	PID1	INTRON	0.02859706
105	104	rs475738	FGF20	3DOWN	0.028540467

- The 160 SNPs are in or near to 140 genes
- One SNP (rs10496476) is located within the intron of gene DPP10, which has been shown to be associated with asthma in multiple populations, based on a recent review
- Most genes are novel

Significance of the Research

Clinical Utility

By identifying **specific genetic markers** (SNPs) linked to exacerbation risk, we move closer to **Personalized Medicine**. This enables earlier therapeutic interventions for high-risk children, potentially reducing hospitalizations.

Methodological Innovation

We successfully demonstrated that **Random Forests** can handle the "Small N, Large P" problem in genomics. This validated a new approach for integrating massive genetic datasets with clinical variables without overfitting.

Biological Insight

The model identified novel genetic targets (like **DPP10**) by capturing non-linear gene-gene interactions that traditional linear models miss, opening new avenues for understanding asthma pathology.

Recognition

ATS 2009: Featured at the "Scientific Breakthroughs of the Year" symposium.

Impact: Categorized as "Highly Accessed" (1,120 accesses 4 months after publication) by BMC Medical Genetics, influencing future ML-GWAS integration studies.

Selected Publications: GWAS & NGS

Genome-Wide Association Studies (GWAS)

- Handedness GWAS (Meta-analysis) — Nat Hum Behav (2020)
- Basal Cell Carcinoma — PLoS One (2011)
- Nevus Count / Melanoma Risk (NID1) — Hum Mol Genet (2011, Co-first)
- Cutaneous BCC & SCC — Hum Mol Genet (2011)
- Pancreatic Adenocarcinoma Survival — Gut (2012)
- Severe Asthma Exacerbation (RF GWAS) — BMC Med Genet (2011)
- eQTL Mapping — Hum Mol Genet (2010)
- ABO & Pancreatic Cancer — Cancer Epidemiol Biomarkers Prev (2010)
- Blood Type & Breast Cancer — Int J Cancer (2011)

NGS & Functional Genomics

- Synthetic transcription elongation factors — Science (2017)
- YAP1 → USP31 → NFκB (Sarcoma) — Cancer Research (2018)
- TRIM24 degrader — Nat Chem Biol (2018)
- MYC dependency in ovarian cancer — eLife (2018)
- Super-enhancer BCL2 mechanism — Cancer Cell (2018)
- Warfarin pharmacogenetics — Pharmacogenomics (2015)

ML Expertise & Industry Experience

Functional Genomics & Machine Learning

- ENCODE Pilot Project — Nature (2007)
- Regulatory polymorphisms / eQTL — Genomics (2008)
- Lgl1 regulation during alveolarization — Pediatric Research (2010)
- RF model for asthma exacerbation — BMC Med Genet (2011)
- LSTM Stock Prediction — Kaggle Tutorial (2018)
- Biological information retrieval — ACM CIKM (1998)

Industry Highlights

- Vidagen (Founder): Genetic testing; developed dozens of products; ~ a dozen patents. (Awards: Hundred, Yellow Crane, 3551 Talent Programs).
- Booz Allen Hamilton (Sr. Data Scientist):
  - Genomic data preparation for Million Veteran Program (MVP).
  - Natural Language Processing (NLP) for NIDDK.
- D2M Biotherapeutics: Causal gene identification for cancers and immune diseases (11M+ MR results).
- Head of Biosciences (Singapore): Quantum-physics for biomedical applications.

Vision: Integrating Computational Analysis into BFE

My goal is to establish a functional lab at GTIIT that bridges the gap between biological engineering and data science.

Curriculum

Introducing practical Machine Learning and Bioinformatics training specifically designed for BFE students to modernize their skillset.

Research

Developing explainable AI pipelines for complex food and biological data, enabling data-driven discoveries in biotech.

Innovation

Translating large-scale "Big Data" into actionable engineering insights for industrial applications.

Family

Thank You

Questions & Discussion

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Backup Slides

Recall Sensitivity True positive rate (TPR)	$\frac{TP}{FN+TP} = \frac{TP}{P}$
False positive rate (FPR) False alarm rate	$\frac{FP}{TN+FP} = \frac{FP}{N}$
Specificity True negative rate (TNR)	$\frac{TN}{TN+FP} = \frac{TN}{N} = 1 - FPR$
Precision	$\frac{TP}{TP+FP}$
False negative rate (FNR)	$\frac{FN}{FN+TP} = \frac{FN}{P}$
Accuracy	$\frac{TP+TN}{P+N} = \frac{TP+TN}{TP+TN+FP+FN}$

		Predicted	
		Positive	Negative
Actual	Positive	TP	FN
	Negative	FP	TN

Receiver-operating characteristic curve (ROC)

$$\boxed{FDR = \frac{FP}{FP+TP}}$$

Table 1 Sample Characteristics (All Subjects Are Caucasian)

	Training Population N = 417		Testing Population N = 164	
	Exac.	Non-exac.	Exac.	Non-exac.
Subjects	127 (30%)	290 (70%)	50 (30%)	114 (70%)
Age (mean ± s.d.)	8.41 ± 2.07	8.89 ± 2.07	8.54 ± 2.28	9.45 ± 2.19
Male	69%	61%	46%	53%
FEV1% (mean ± s.d.)	92.9 ± 15.7	93.5 ± 13.2	95.4 ± 17.1	95.4 ± 13.5
Treatment				
Budesonide	20%	31%	36%	32%
Nedocromil	28%	29%	30%	32%
Placebo	52%	39%	34%	36%